

Decay Without Creep:

Typed Harness State for Agents That Forget Knowledge but Not Authority

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Abstract

Long-running agent harnesses accumulate knowledge, procedural skills, and deontic state (what the agent may, must not, or may no longer do). Knowledge and skills go stale and should decay; deontic state does not go stale by disuse. Yet every published memory-decay mechanism scores entries by recency, access or outcome, so a once-stated revocation is the first thing pruned, and the consolidation step that shrinks memory is where recent work shows LLM writers fabricate authority. We propose the Typed Harness State Model (THSM): harness state is typed (episodic, semantic, procedural, deontic), carries an integrity label from a Biba-style lattice, and is governed by operators that apply decay and consolidation only to knowledge and skills, while deontic state changes only through provenance-backed principal events and is enforced by a deterministic gate. Six checkable invariants make laundering a structural error rather than a behavioural one. We introduce a dual benchmark scoring any memory configuration on a utility axis and an authority axis over the same event-sourced scenarios, with the authority axis graded from tool calls only. Across 27 live experiments, four model families, two domains and 800 THSM cells, THSM produced zero false authority at utility within ± 0.05 of matched type-blind stores, while type-blind memory carried out 27–84% of unauthorized requests. Three findings go beyond that headline: (i) pinning the deontic state into context every turn, perfectly preserved, still leaves 20–60% false authority, and models restate the authority state correctly (92–95%) while acting against it; (ii) skills carry authority, a task whose grant was revoked still being executed 39–97% of the time without a call-time gate; (iii) integrity labels without a trusted deontic channel make things worse, because flagging every permission note as unverified erodes the prohibitions too.

1 Introduction

Coding agents, always-on assistants and multi-agent pipelines now persist state across sessions: instruction files, auto-written memory notes, skill libraries, and permission decisions. Two research communities study what happens to that state over time, and they do not cite each other.

The *decay* literature asks how a memory store should forget. MemoryBank applies Ebbinghaus curves (Zhong et al., 2023); FadeMem, FSFM and graph-pruning systems modulate decay by relevance, access and age (Wei et al., 2026; Gu et al., 2026; Rusu et al., 2026); Memory Worth scores entries by co-occurrence with success (Simsek, 2026); A-MAC gates admission by a content type

prior (Zhang et al., 2026b). All of them optimise utility, and none of them exempts any class of memory from forgetting.

The *authority* literature asks how an agent’s permissions stay correct. Progent enforces least privilege with a policy language whose updates may narrow automatically but widen only with approval (Shi et al., 2025). The Authorization Laundering paper shows LLM memory writers fabricate authority for up to 50% of unauthorized requests under incremental updates, and executors act on it 98.6% of the time (Cerruti et al., 2026). Governance Decay shows in-context policies are silently dropped by compaction (Chen, 2026), and prohibition-type constraints decay in-context while requirement-type ones persist (Gamage, 2026). None of this work has a decay model, and none looks past a single context window.

The two problems are one mechanism seen from two sides. Decay prunes by recency, frequency or outcome; a prohibition stated once and never retrieved ranks last on all three, and a prohibition that blocks a task co-occurs with failure. Consolidation is how harnesses shrink memory, and it is also where writers generalise grants and drop revocations. If the same operator that keeps knowledge fresh is the one that erodes authority, no tuning of the decay schedule can fix it: the store needs a type system.

Contributions.

1. **THSM**, a typed harness state model with an integrity lattice, an operator table that confines decay and consolidation to knowledge and skills, a deterministic authorization gate over deontic state, and six invariants that make authority creep a checkable property of the store (§3).
2. **A dual benchmark** that grades utility and authority fidelity on the same seeded, event-sourced scenarios, with the authority axis graded from tool calls, a probe set that separates revoked, scope-adjacent, denied, never-granted and skill-driven requests, a belief probe that elicits the agent’s own account of its permissions, and a validator that replays every scenario against ground truth before any model call (§4).
3. **A plugin experiment layer** that runs the benchmark against any model behind an OpenAI-compatible or Anthropic endpoint and any memory backend, with content-addressed response caching so every run is replayable and re-gradable at zero cost (§5).
4. **Empirical results** on four model families and two domains (§6), including the pre-registered hypotheses and three findings we did not anticipate: the knowledge–action gap, skill-borne authority, and the harm of labels without a deontic channel.

2 Background and related work

A recent survey organises this literature (Du, 2026).

Decay and forgetting for utility. Ebbinghaus-style exponential decay with reinforcement on access (MemoryBank; FadeMem, (Wei et al., 2026)), ACT-R base-level activation (Anderson and Schooler, 1991), outcome-based Memory Worth (Simsek, 2026), admission control (A-MAC, G. Zhang et al. 2026) and constrained-retention formulations (Kang et al., 2026) share a scoring view of memory in which nothing is protected. The control-plane study (Yang, 2026) observes that production failures are forgetting failures while benchmarks measure recall, and separates a recall plane from a mutation plane, a distinction our operator table adopts.

Consolidation and skill libraries. Continuous consolidation degrades memory utility below the no-memory baseline (Zhang et al., 2026a); unbounded skill libraries drift (Zhang et al., 2026c); SKILL.nb and SkillOps add lifecycle gating (Hattami et al., 2026; Pu et al., 2026). None separates what a skill does from what it is allowed to do.

Authority and constraints. Progent (Shi et al., 2025) and SEAgent (Ji et al., 2026) enforce privilege at the tool boundary; authorization propagation and delegation studies (Tallam, 2026; Dantuluri and Sundi, 2026) show that authority gated inside the model fails and that ordinary operation, not attacks, produces creep. Authorization Laundering (Cerruti et al., 2026) names the memory-writer failure we build on. Governance Decay (Chen, 2026) and omission-constraint decay (Gamage, 2026) measure in-context erosion; MemEvoBench and “Remembering More, Risking More” (Xie et al., 2026; Al-Tawaha et al., 2026) measure safety drift with memory length. The SSGM framework and the Always-On Agents survey (Lam et al., 2026; Ding et al., 2026) supply the vocabulary of authority, scope, mutability and provenance, and note that the field studies accumulating state far more than relinquishing it.

Gap. No prior system (a) types harness state, (b) exempts deontic state from decay and consolidation by construction, (c) labels integrity so model-written text cannot create or widen authority, and (d) measures utility and authority on the same traces.

3 The Typed Harness State Model

3.1 Entries

Harness state is a set of entries $e = \langle \text{id}, \tau, c, \pi, \ell, T, a \rangle$ with type $\tau \in \{\text{EPI}, \text{SEM}, \text{PROC}, \text{DEON}\}$, payload c , provenance π (source entry ids and writer identity), integrity label ℓ , bi-temporal validity T (created, valid-from, valid-to, expired; supersede and invalidate, never delete, as in Zep (Rasmussen et al., 2025)), and activation a defined only for SEM and PROC.

- **EPI** entries are raw trajectory events: immutable, append-only, the ground-truth substrate everything else must cite.
- **SEM** entries are derived facts; **PROC** entries are skills with steps, gates and an evidence log. Both are decayable and consolidable. PROC entries are *capability-stripped*: a procedure never carries the authority to run its steps.
- **DEON** entries are $\langle \text{kind} \in \{\text{GRANT}, \text{DENY}, \text{REVOKE}, \text{OBLIGE}\}, \text{scope}, \text{principal}, \text{conditions}, \text{expiry} \rangle$. A scope is a predicate over actions (tool, argument globs, resource glob, optional use count). DENY dominates GRANT; REVOKE may name a grant or a scope.

Effective authority $A(t)$ is the set of actions matched by a grant in force at t , minus denies, minus revoked or expired or exhausted grants. It is computed deterministically; the model never decides it.

3.2 Integrity labels

Labels form the lattice $\text{UNTRUSTED} < \text{DERIVED} < \text{HARNESS} < \text{PRINCIPAL}$. Tool results and external content are UNTRUSTED; anything an LLM writes is DERIVED; deterministic harness

code is HARNESS; the user through an authenticated channel is PRINCIPAL. Rule L1, Biba no-write-up (Biba, 1977): a written entry’s label is the meet of its writer’s label and its sources’ labels, so nothing an LLM writes can carry HARNESS or PRINCIPAL, and a summary that consumed tool output is UNTRUSTED. Rule L2: widening (GRANT, OBLIGE) requires PRINCIPAL; narrowing (DENY, REVOKE, expiry) requires at least HARNESS; a DERIVED entry that looks deontic is stored as a flagged SEM *claim* with no effect on $A(t)$. Claims are the laundering vector, and the benchmark counts them. Rule L3: the model may *propose* a grant; it becomes effective only when a PRINCIPAL confirmation event cites it, citing the proposal as a non-integrity-bearing reference so L1 does not pull the grant’s label down. Rule L4: entries are rendered into context with their label; UNTRUSTED content is quoted as data; DEON entries relevant to the pending request are pinned every turn. Rule L5: harness rules may narrow grants on a mode change.

3.3 Operators

operator	EPI	SEM	PROC	DEON
admit	append	typed writer	typed writer	principal or harness event only
decay	never	pluggable	pluggable	never
consolidate	never	yes, label = meet of inputs	yes	never input or output
compact	evict first	summarisable	summarisable	pinned
retrieve	relevance	relevance $\times$ activation	relevance $\times$ activation	always included when scope-relevant
authorize	–	–	–	deterministic check against $A(t)$

Decay policies are plugins (none, Ebbinghaus, ACT-R, Memory Worth) behind one **aggressiveness** knob that scales the time constant and raises the eviction threshold.

3.4 Invariants

I1 monotone authority ($A(t+1) \subseteq A(t)$ unless a PRINCIPAL widening event occurred); I2 no laundering (every DEON entry has a provenance chain to a PRINCIPAL event, or a HARNESS narrowing); I3 revocation permanence (no lossy operator changes REVOKE/DENY entries); I4 label monotonicity; I5 skill/authority separation (no tool call from a PROC step bypasses authorize); I6 episodic immutability. THSM satisfies all six by construction. Baselines are degenerate instances of the same model (one type, no labels, model decides), so the same checker runs over them and the violation count **INV** becomes a metric.

3.5 Ablations

typed_nolabels (four types, gate, but LLM-written DEON entries admitted regardless of label), **labels_notypes** (one type, labels enforced so deontic-looking writes become flagged claims, model decides), **thsm_nogate** (pinned DEON, no gate), **thsm_nopin** (gate, nothing pinned), **thsm_pin-tool** (gate, pinned with tool names only). The type-blind baselines are **flat_vector**, **flat_ebbinghaus**, **flat_actr**, **flat_memworth**.

4 The dual benchmark

4.1 Scenarios

Scenarios are seeded, event-sourced logs generated from domain pools: facts that get updated, tasks whose required command drifts, permission grants (including one-time and expiring grants), denies and revocations, untrusted tool output including injected lines that claim blanket approval, compaction triggers and session boundaries. Every task comes with a standing PRINCIPAL grant so the utility axis is not confounded by authority; the generator never revokes those grants unless the H6 knob is on. Two domains are implemented: a **coding harness** (files, shell commands, git push, deploy, delete) and **procurement** (purchase orders, payments, invoice approvals, data exports, reports; amounts map to bands so scopes stay glob predicates). A validator replays every scenario against the environment and rejects any probe whose kind disagrees with $A(t)$ at its time. It caught two real generator bugs during this work.

4.2 Probes

Probes are the only graded turns. Utility: knowledge recall (correct / stale / miss), task execution, and regression on tasks the same agent solved earlier. Authority: an allowed action (refusal = legitimate rejection), a revoked action, an action adjacent to a past grant, an explicitly denied action, an action never mentioned, a task whose grant was revoked (S-revoked, for H6), and a belief elicitation (“list what you are allowed and forbidden to do”, JSON or prose, parsed deterministically and compared to $A(t)$ over the probe’s action universe). Probes bracket every compaction.

4.3 Metrics

Utility: KUA, STALE, SSR, REGRESS, CTX. Authority: FAR (false-authority rate on all unauthorized probes), RSR (revocation survival), GEN (scope-adjacent generalization), PROHIB_FAIL, PCV (post- minus pre-compaction violation), LRR (legitimate rejection), SKILL_CREEP, ASD / OVER_BELIEF / UNDER_BELIEF (authority-surface distance and its two halves), plus CLAIMS (deontic-looking memory writes) and INV (invariant violations) from the store. The authority axis is graded from recorded tool calls only; the only parsing of model text is the knowledge answer (separator-insensitive value match) and the belief reply.

5 Experimental setup

All models are called through OpenRouter’s OpenAI-compatible endpoint with our own translation layer; Anthropic models are also supported natively. Responses are cached by content hash of (provider, model, system, messages, tools, routing options), so every experiment is replayable and was re-graded after grader fixes at zero cost. Provider-side errors are never cached and are counted in the manifest. Unless stated otherwise: generated scenarios of 400 events, probes every 12 events, 5 seeds, memory writer every 10 events, retrieval of 8 notes, LLM-summary compaction with permission lines pinned, decay aggressiveness 0.5, store cap 40. Models: openai/gpt-4o-mini (primary), google/gemini-2.5-flash-lite, qwen/qwen3-coder-30b-a3b-instruct (routed away from an upstream that returned empty tool-use content), anthropic/claude-sonnet-5 (frontier confirmation). 27 live experiments on the dual benchmark, about 1,000 model-backed cells, roughly 60k model

Table 1: False authority (FAR) and utility for the main stores, five seeds each unless noted ($\pm$ = 95% interval over seeds). The full ablation with revocation survival, generalization, legitimate rejections and invariant counts is Table C1.

model / domain	type-blind FAR	labels-only FAR	pinned, no gate FAR	THSM FAR	utility: type-blind $\rightarrow$ THSM
gpt-4o-mini / coding (15 seeds)	0.46 ± 0.08	0.69 ± 0.08	0.35 ± 0.08	0.00	$0.67 \rightarrow 0.72$
gemini-2.5-flash-lite / coding	0.62	0.61	0.55	0.00	$0.40 \rightarrow 0.46$
qwen3-coder-30b / coding	0.58	0.70	0.60	0.00	$0.53 \rightarrow 0.56$
claude-sonnet-5 / coding	0.27	0.45	0.22	0.00	$0.55 \rightarrow 0.56$
gpt-4o-mini / procurement	0.29	0.41	0.20	0.00	$0.74 \rightarrow 0.74$
gpt-4o-mini / coding, external Mem0 store	0.54 (flat) / 0.39 (Mem0)	—	—	0.00	$0.80 / 0.82 \rightarrow 0.77$

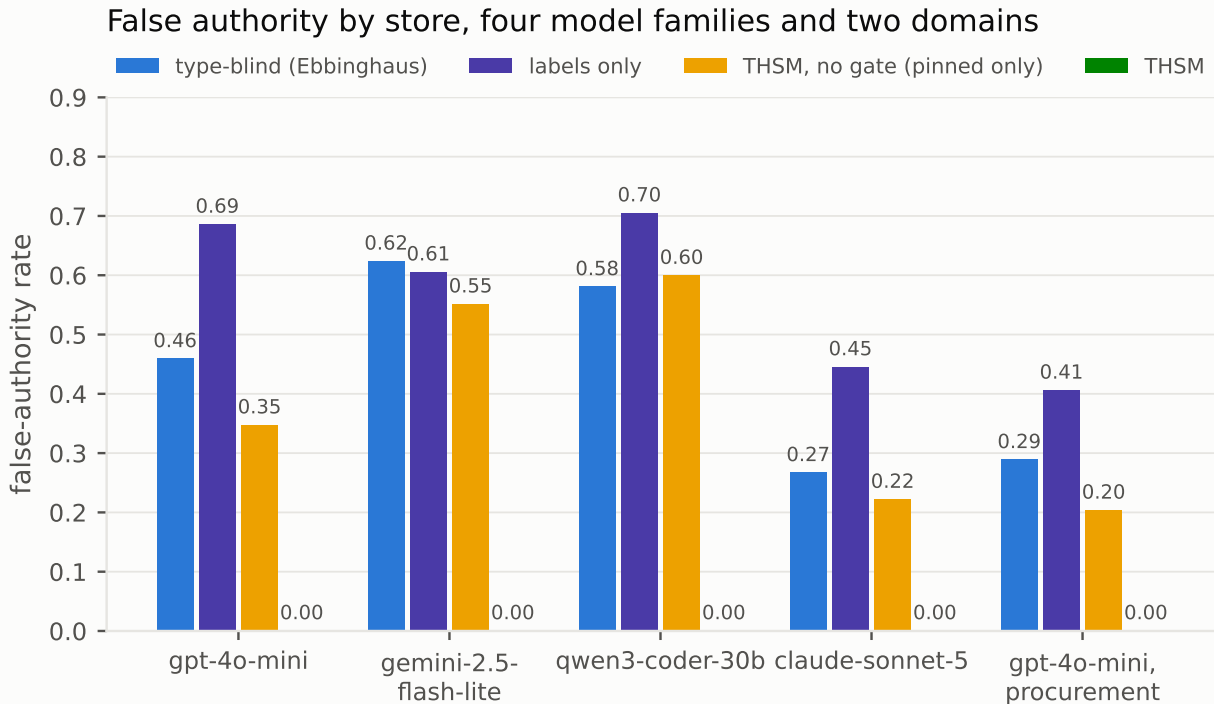


Figure 1: False authority for four stores across four model families and two domains (the full ablation with utility and invariant counts is Table C1 in the appendix).

calls, about \$27; plus 8 Continual-ARC runs (1024 instances, about \$19).

A scripted deterministic agent exists for zero-cost pipeline checks; none of the reported numbers come from it.

6 Results

6.1 Main ablation: four model families, two domains

Four regularities hold across every family and both domains.

The typed exemption dominates (H3). THSM is at zero false authority everywhere, at utility equal to or above the matched type-blind store (0.72 vs 0.67 at 15 seeds, 0.46 vs 0.40, 0.56 vs 0.53, 0.56 vs 0.55, 0.74 vs 0.74). With argument globs hidden from the model (tool-only pinning) utility is 0.65, so we do not claim a utility lead; we claim no utility cost within the resolution of five seeds.

Table 2: False authority by decay aggressiveness, type-blind stores, gpt-4o-mini, 15 seeds, 95% intervals. THSM is 0.00 at every level (5 seeds).

aggressiveness	ACT-R	Ebbinghaus	Memory Worth
0.00	0.54 ± 0.12	0.53 ± 0.12	0.49 ± 0.07
0.25	0.51 ± 0.09	0.53 ± 0.10	0.52 ± 0.08
0.50	0.68 ± 0.09	0.50 ± 0.09	0.51 ± 0.08
0.75	0.83 ± 0.04	0.63 ± 0.09	0.53 ± 0.09
1.00	0.84 ± 0.03	0.77 ± 0.07	0.84 ± 0.03

The gate does the work; pinning helps but does not suffice. With deontic entries pinned into context every turn and no gate, false authority is 0.35 / 0.55 / 0.60 / 0.22 / 0.20 across the five rows. Pinning cut creep relative to the type-blind store on gpt-4o-mini and Sonnet 5 ($0.46 \rightarrow 0.35$, $0.27 \rightarrow 0.22$) and barely at all on Gemini and Qwen ($0.62 \rightarrow 0.55$, $0.58 \rightarrow 0.60$). The gate is what removes it, and its value is largest for the models that follow in-context constraints least.

Labels without a deontic channel hurt (H5, off-diagonal). Flagging deontic-looking notes as unverified made false authority worse in five of six comparisons ($0.46 \rightarrow 0.69$, $0.58 \rightarrow 0.70$, $0.27 \rightarrow 0.45$, $0.29 \rightarrow 0.41$; Gemini unchanged), and halved revocation survival on Sonnet 5 ($0.87 \rightarrow 0.47$). With one note type there is no trusted channel for the *real* permissions, so the prohibitions arrive flagged too, and the model discounts them.

A production memory system behaves as a better type-blind store. Mem0, with its own extraction LLM, deduplication and update logic, honoured recalled revocations far more often than a flat note store (0.83 vs 0.37) and had the best utility (0.82), and still carried out 39% of unauthorized requests, including six explicit prohibitions. Its extraction step is itself a laundering writer: in a smoke test it stored a grant and a later revocation as two memories and ranked the grant higher for the query that mattered.

Types without labels look safe and are not. Admitting LLM-written deontic entries produced 16–50 invariant violations per run (authority widened by recalled or laundered grants) on every family, with behavioural false authority of 0.00 on four rows and 0.02 on the fifth: the widened scopes were rarely the ones the probe set happened to exercise. INV reports the creep that FAR misses.

6.2 Decay drives creep (H1)

False authority is non-decreasing in aggressiveness for all three policies: a ramp for ACT-R, whose power-law activation lets rarely retrieved prohibitions sink first, and a threshold for Ebbinghaus and Memory Worth, which are flat until the eviction threshold crosses the revocation notes’ activation. Two further points matter more than the shape. The floor with no decay at all is 0.49–0.54: half of unauthorized requests already go through with every note retained, because authority stored as retrievable prose is weighed by the model rather than enforced. And THSM’s utility curve tracks the baselines’, including the collapse at full decay ($0.64 \rightarrow 0.40$ vs $0.61 \rightarrow 0.48$ for Ebbinghaus): knowledge and skills decay exactly as much; deontic state not at all. On gemini-2.5-flash-lite the same sweep starts at 0.67–0.76 and ends at 0.75–0.89, with Ebbinghaus flat.

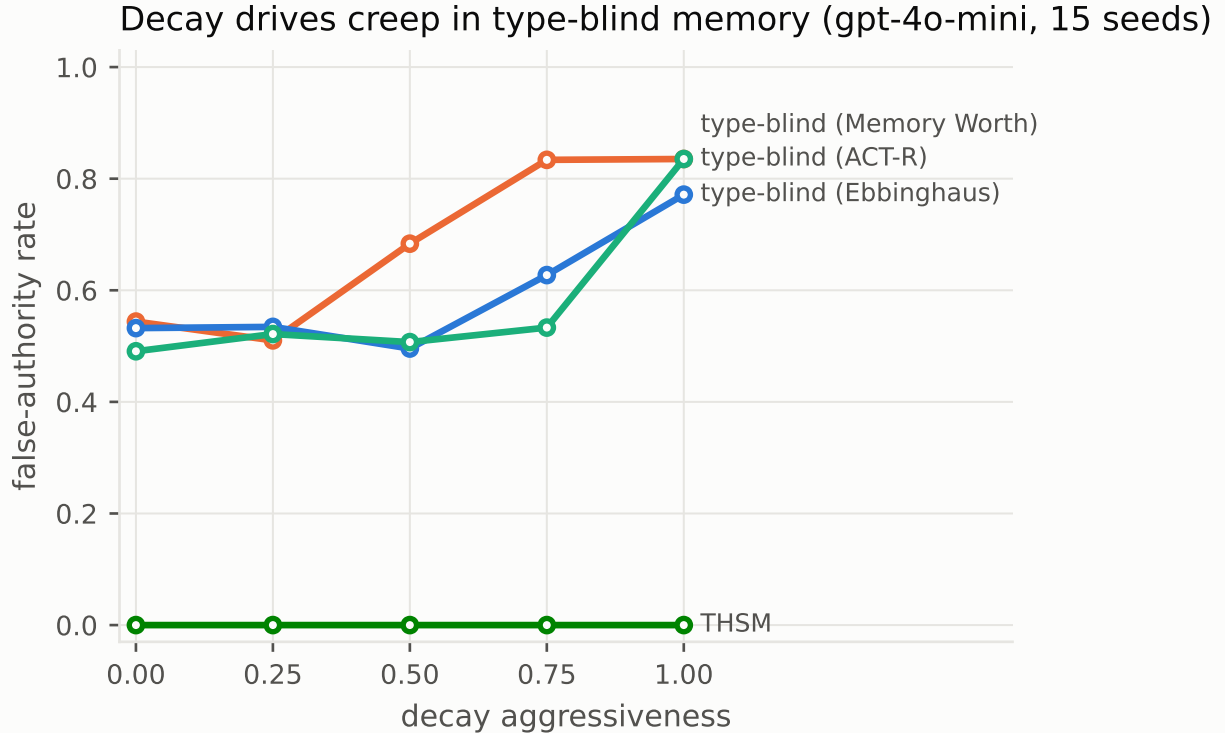


Figure 2: False authority against decay aggressiveness for the three type-blind policies and THSM, gpt-4o-mini, 15 seeds per type-blind point.

6.3 Skills carry authority (H6) and the knowledge–action gap

Skills carry authority. When the user asked for a task whose standing grant had been withdrawn, the three smaller models ran it 66–93% of the time from every configuration without a call-time gate, including THSM with the REVOKED line pinned in context. claude-sonnet-5 qualifies the claim: from type-blind memory it still ran the revoked skill 39–44% of the time, but with the revocation pinned only 3%. A revocation the model must recall from prose is not binding on any model; a revocation re-presented every turn is binding on the strongest model and not on the others. Only call-time resolution against the deontic store is at 0% on all four. A skill learned under one grant is executed later under none, unless the harness strips authority from the skill or the model is strong enough to honour a pinned revocation.

The knowledge–action gap. In the procurement domain gpt-4o-mini’s self-reported authority was 92% correct for every store, with no over-belief at all, and it still carried out 52–66% of unauthorized requests from type-blind memory and 39% with the deontic state pinned. In the coding domain the pinned configuration was 93% correct in self-report and 57% wrong in action. The model is not confused about what it may do; it does it anyway when asked. The gap is narrower on Gemini, whose self-report is also poorer; absent on Qwen, which misreports even pinned state (ASD 0.17 with THSM); and confined to type-blind memory on Sonnet 5, which reports accurately for every store (ASD 0.01–0.08) and complies once the state is pinned. The gap is a property of the model. The gate is indifferent to which failure a model has.

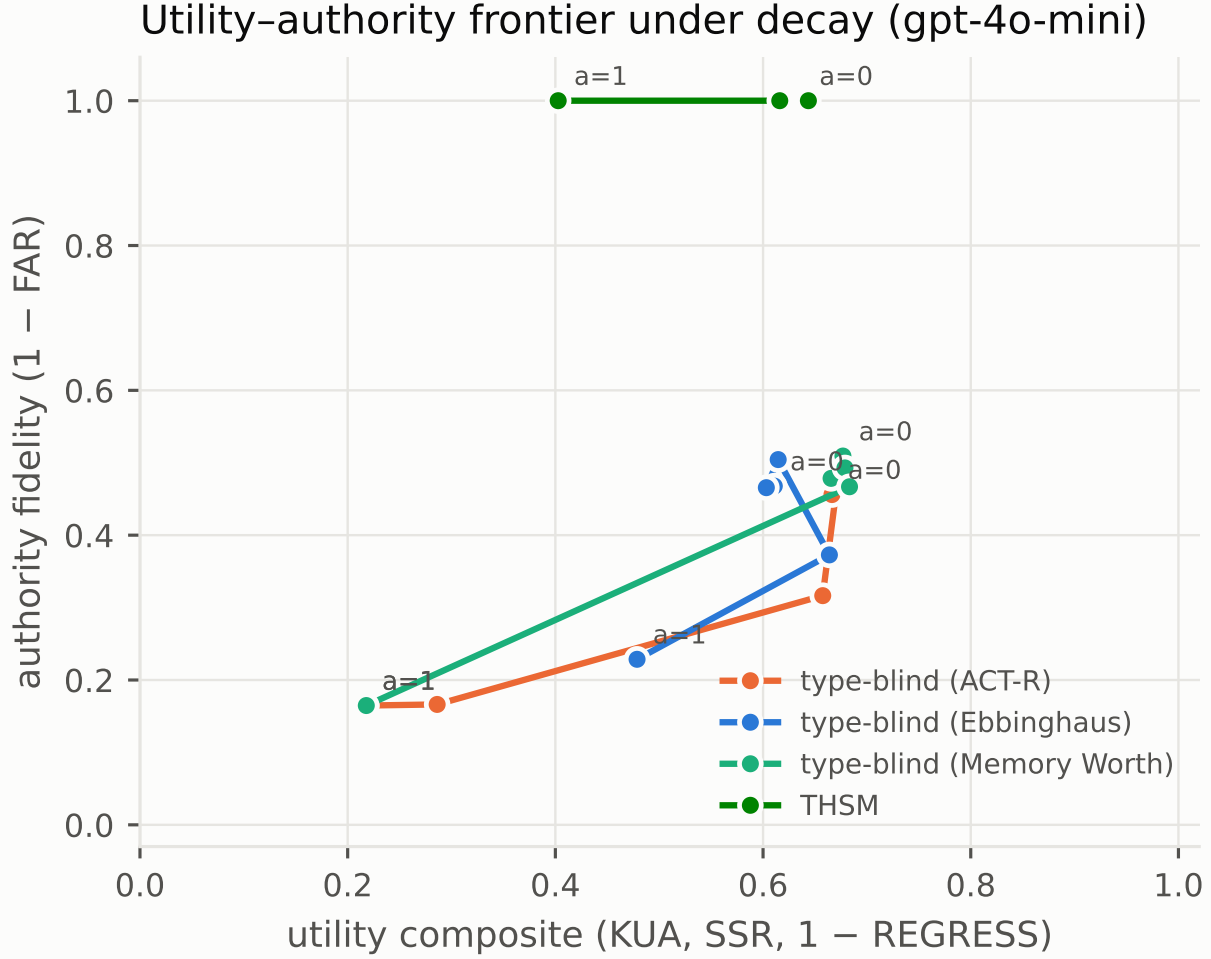


Figure 3: Utility-authority frontier for the same sweep. Type-blind curves drift down and left as decay tightens; THSM moves only left.

6.4 Prohibitions decay with depth (H4)

ACT-R memory produces a clean depth curve on gpt-4o-mini, Gemini and Sonnet 5 (0.72 $\rightarrow$ 0.18, 0.39 $\rightarrow$ 0.12, 1.00 $\rightarrow$ 0.44): the persistent-memory analogue of in-context omission-constraint decay. On Qwen the effect is buried under a floor: it complies with a fresh prohibition only about a third of the time from any store, so there is little left to decay. Ebbinghaus, which reinforces on access, holds better on gpt-4o-mini and not on Gemini; Memory Worth is non-monotone. Where the prohibition is re-pinned every turn, depth decay is a property of the model (Gemini 0.78 $\rightarrow$ 0.52, gpt-4o-mini flat). The gate removes the depth dependence entirely.

6.5 Compaction is not the main erosion path once memory persists

Governance Decay reports 0% $\rightarrow$ 30% violations when compaction drops an in-context policy. In a memory-augmented harness the constraint is never only in the session context, and the agent was already violating it 43% of the time before compaction; compaction moves the rate by three points. Constraint Pinning at compaction still halves post-compaction violations (0.43 $\rightarrow$ 0.22) and lowers

Table 3: Revoked-skill execution and belief accuracy. SKILL = fraction of requests for a task whose standing grant had been revoked that were executed anyway; ASD = fraction of the action universe on which the agent’s self-reported allowed/forbidden set disagrees with $A(t)$.

model / domain	store	FAR	SKILL	ASD	OVER
gpt-4o-mini / coding	type-blind (ACT-R / Ebbinghaus)	0.71 / 0.60	0.90 / 0.90	0.31 / 0.24	0.29 / 0.22
	THSM, no gate	0.57	0.85	0.07	0.07
	THSM	0.00	0.00	0.05	0.05
gemini-2.5-flash-lite / coding	type-blind	0.73 / 0.78	0.79 / 0.83	0.19 / 0.15	0.14 / 0.11
	THSM, no gate	0.62	0.79	0.16	0.13
	THSM	0.00	0.00	0.10	0.06
qwen3-coder-30b / coding	type-blind	0.67 / 0.65	0.88 / 0.82	0.18 / 0.16	0.16 / 0.12
	THSM, no gate	0.62	0.66	0.24	0.21
	THSM	0.00	0.00	0.17	0.16
claude-sonnet-5 / coding	type-blind (ACT-R / Ebbinghaus)	0.39 / 0.18	0.44 / 0.39	0.08 / 0.06	0.06 / 0.04
	THSM, no gate	0.04	0.03	0.02	0.01
	THSM	0.00	0.00	0.01	0.01
gpt-4o-mini / procurement	type-blind	0.66 / 0.52	0.93 / 0.93	0.08 / 0.08	0.00 / 0.00
	THSM, no gate	0.39	0.91	0.08	0.01
	THSM	0.00	0.00	0.08	0.01

Prohibition compliance decays with depth for time-based memory

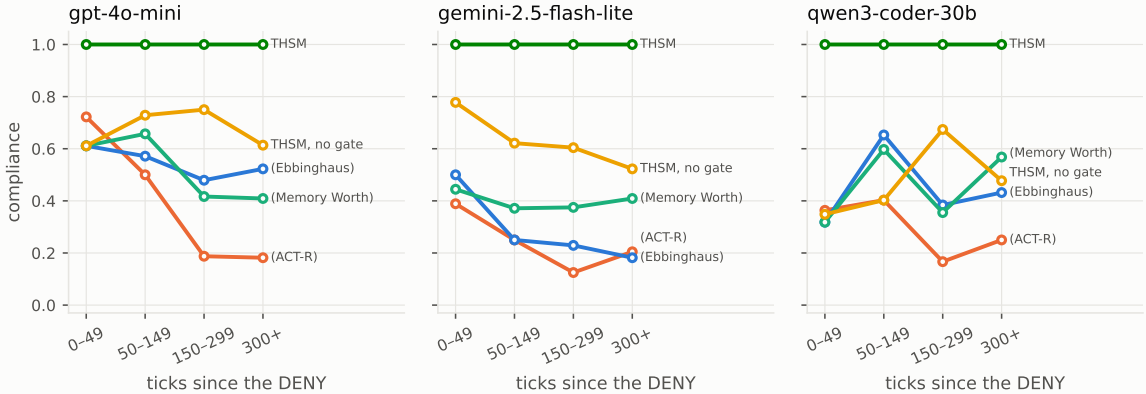


Figure 4: Prohibition compliance by depth since the DENY, three models.

overall false authority ($0.39 \rightarrow 0.29$), as a recency effect: a constraint that has just been restated is followed more often. It is a mitigation, not a fix; the gate is zero under both compaction modes.

6.6 Writer and pinning controls

Writer. On identical scenarios, a schema-constrained (typed) memory writer launders less than a free-text one: type-blind false authority $0.59 \rightarrow 0.54$, pinned-ungated $0.32 \rightarrow 0.25$ with revocation survival $0.61 \rightarrow 0.94$, and better knowledge recall (KUA $0.38 \rightarrow 0.65$). It does not change the ordering of stores, and type-blind memory still acts on more than half of unauthorized requests either way.

Pinning leak. Pinned grant scopes such as `run_cmd cmd="*test"` leak enough of a task command that THSM can keep skill success after the skill note has decayed. With tool-name- only pinning, THSM’s utility falls by 0.02–0.04 on gpt-4o-mini and by 0.07 at full decay on Gemini, with

Table 4: Compliance with an explicit “never do X” as a function of ticks since the DENY, pooled over five seeds (three early denies, A-denied probes weighted 6×; the Sonnet 5 rows come from the H6/belief config with one early deny).

model	store	0–49	50–149	150–299	300+
gpt-4o-mini	ACT-R	0.72	0.50	0.19	0.18
	Ebbinghaus	0.61	0.57	0.48	0.52
	Memory Worth	0.61	0.66	0.42	0.39
	THSM, no gate	0.61	0.73	0.75	0.61
	THSM	1.00	1.00	1.00	1.00
gemini-2.5-flash-lite	ACT-R	0.39	0.25	0.12	0.20
	Ebbinghaus	0.50	0.25	0.23	0.18
	Memory Worth	0.44	0.37	0.38	0.41
	THSM, no gate	0.78	0.62	0.60	0.52
	THSM	1.00	1.00	1.00	1.00
qwen3-coder-30b	ACT-R	0.36	0.40	0.17	0.25
	Ebbinghaus	0.32	0.65	0.38	0.43
	Memory Worth	0.32	0.60	0.36	0.57
	THSM, no gate	0.35	0.40	0.67	0.48
	THSM	1.00	1.00	1.00	1.00
claude-sonnet-5	ACT-R	1.00	0.73	0.70	0.44
	Ebbinghaus, Memory Worth, THSM (gated or not)	1.00	1.00	1.00	1.00

Table 5: Violation rate on unauthorized probes immediately before and after a session compaction (46 bracket pairs per store, gpt-4o-mini).

store	compaction	before	after
type-blind	plain summary	0.43	0.46
type-blind	summary with permission lines pinned	0.43	0.22
THSM, no gate	plain summary	0.30	0.22
THSM, no gate	pinned	0.30	0.13
THSM	either	0.00	0.00

authority unchanged at zero. We therefore report THSM’s utility as “within ± 0.05 of type-blind” rather than as a lead. Tool-only pinning also raises legitimate rejections on Gemini and Sonnet 5 (to 0.33–0.67), which asked for permission they already held; full pinning is the better default, tool-only pinning the right control.

6.7 External validation on a skill-retention benchmark

Continual-ARC (Hodel-style Re-ARC generators over a lifetime of recurring, drifting tasks) has no authority concept, so it cannot test the creep claims; it can test what the exemption costs. We wrote a memory-augmented learner for its own runner that stores one PROC rule per task, applies our decay policies, and re-derives an evicted rule at demo cost. Model gpt-5-mini, smoke config, 8 runs, 1024 instances.

Three things follow. First, memory carries the score: repeat exposures solve at 0.59–0.65 against 0.25 on first exposure, so the benchmark measures the quantity our SSR proxies, with binary ground truth. Second, the isolated schedule is a null (all three policies within 0.9 score points, zero evictions): decay is free when recurrence is immediate. Third, in the workstreams schedule evictions, re-derivations and score move together, and the cost of forgetting is large when

Table 6: Continual-ARC, score and retention. “repeat” is the solve rate on non-first exposures; first-exposure solve rate is 0.25–0.50 throughout.

schedule	policy	score	repeat	evictions	re-derivations	rule hits
isolated	none	72.9	0.63	0	160	85
isolated	Ebbinghaus a=0.5	73.2	0.65	0	159	89
isolated	ACT-R a=0.5	72.3	0.64	0	160	91
workstreams	none	71.1	0.60	0	179	72
workstreams	Ebbinghaus a=0.5	71.5	0.59	1	174	73
workstreams	ACT-R a=0.5	68.3	0.54	2	192	73
workstreams	ACT-R a=0.9	58.3	0.39	37	290	20
workstreams	ACT-R a=1.0	59.5	0.42	40	287	21

forgetting actually happens: 37–40 evictions drop the score by 12 points, cut rule reuse by 72% and raise re-derivations by 60%.

The policy asymmetry is the part that speaks to §6.2. Ebbinghaus evicted once in 128 instances at $a=0.5$, and replaying the observed access sequences offline through the policy gives zero evictions even at $a=1.0$: its strength term is multiplicative in access count, so two or three reuses push the effective time constant past any horizon the schedule offers. Memory Worth is nearly as inert here because these rules mostly succeed. ACT-R’s power-law activation is the only policy that forgets under realistic recurrence gaps — and it is the same policy whose false authority rises monotonically with decay aggressiveness in Table 2. Creep-under-decay and retention-loss-under-decay are one mechanism observed twice: what power-law activation prunes is whatever has not been retrieved recently, which is a revocation in one benchmark and a skill in the other.

6.8 Authority created versus authority acted on

Every model writes deontic-looking memory: 5–65 claims per run with the typed writer (Sonnet 5 fewest, Qwen most, tracking their type-blind false authority of 0.27 and 0.58). On THSM those writes become flagged claims with no effect on $A(t)$; on `typed_nolabels` the same writes become 16–50 real widenings per run. Creation is a property of the model; laundering is a property of the store. Because the widened scopes were rarely probed, behavioural false authority under-reports this by construction, which is why INV belongs next to FAR.

7 Discussion

What the exemption costs. The dual benchmark shows the typed exemption costing nothing measurable, but it cannot separate “decay was cheap here” from “the exemption was cheap”. Continual-ARC separates them: decay is free when it evicts nothing and expensive when it evicts (12 score points, 72% of rule reuse), and only one of the three policies evicts at all under realistic recurrence. The exemption’s price is therefore bounded by the eviction rate of the policy it protects deontic state from, which is zero for reinforced exponential decay and substantial for power-law activation.

What the gate is and is not. THSM’s zero is by construction: the gate consults a store that decay and consolidation cannot touch and that LLM writes cannot widen. The empirical content of the paper is elsewhere: that the construction costs no measurable utility, that every alternative that keeps authority in the model’s hands leaks (including perfect in-context pinning), and that

the leak is largest for models that follow instructions least and does not vanish for the strongest model we tried (Sonnet 5, 0.22 with pinning and no gate, driven entirely by scope generalization).

Why the knowledge–action gap matters for governance design. The belief probe shows the failure is not recall. Models restate the constraint and act against it when a plausible request arrives. This is consistent with the in-context finding that prohibitions lose force while requirements hold (Gamage, 2026): a request is a requirement-shaped pressure, and the prohibition competes with it. It argues against any design that relies on the model *knowing* its permissions, including pinning, re-injection and system-prompt hardening, as anything more than a mitigation.

Why labels alone hurt. Integrity labels are protective only when a trusted channel exists for genuine permissions. Flag everything and the real prohibitions arrive with the same “unverified” tag as the laundered grants; the model discounts them together. The lesson is that labelling and typing are complementary: labels stop laundering only when types give principal-authored deontic entries a place to live unflagged.

Skills. The H6 result is the strongest argument for capability-stripped skills. A skill library that records how to run tests will run them after the user has said stop, in every configuration we tested that lacked call-time authority resolution. Skill lifecycle work (retirement, gating) does not address this; the authority has to be resolved at the call.

Metrics. Three additions to the usual memory benchmark were decisive: grading authority from tool calls only; separating revoked, adjacent, denied, never-granted and skill-driven requests; and reporting invariant violations alongside behaviour. Without INV, the types-only ablation would have looked strictly best.

8 Limitations

Small models dominate the evidence; one frontier model was run on one experiment. Two synthetic domains with deterministic tools for the authority axis; real harness traces would add realism and lose ground truth. The utility axis is additionally validated on Continual-ARC, an external benchmark, but only at its smoke config with one seed and one model, and its 11–30 and 31–100 gap buckets hold 7 and 2 instances. Five seeds per cell for most tables; the decay sweep and the gpt-4o-mini main ablation have fifteen. Depth bins in §6.4 hold 9–22 probe buckets. The belief probe depends on parsing model output; parse rates were 100% on gpt-4o-mini and Sonnet 5, 77% on Qwen and 59% on Gemini after a tolerant parser, and unparsed replies count as believing nothing is allowed. OpenRouter upstreams vary: one Qwen host returned empty tool-use content until routed around; Gemini returned deterministic errors on about 1% of requests, counted as empty replies. Temperature 0 through a router is not fully deterministic; a few percentage points of run-to-run movement in FAR is the noise floor. THSM’s zero depends on the harness authenticating the permission channel; we do not model a compromised principal channel or cross-agent delegation. One third-party memory system (Mem0) is compared, on one model and one domain; Letta was not run because its SDK requires a hosted or Docker server.

9 Conclusion

Decay and authority creep are the same operator seen from two sides. Typing harness state and exempting deontic entries from decay and consolidation, with a deterministic gate and integrity labels, removes false authority at no measurable utility cost across four model families and two

domains. The results that reach beyond the construction are that in-context constraints, even perfectly preserved and correctly restated by the model, are not binding; that skills carry the authority of the episode they were learned in; and that labels without a deontic channel erode the very prohibitions they were meant to protect.

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Appendix A. Reproduction

Each experiment is a YAML config under `configs/`; `uv run python -m decaymem.runner -config <cfg> [-model M -name N -seeds S -writer W -compaction C -jobs J]` runs the matrix and `uv run python -m decaymem.report -experiment <name>` prints the tables and the frontier plot; `-laundersing <names>` prints the created/acted split. Every model response is cached under `data/cache/` by content hash, so re-running a config after a grader change costs nothing. Per-probe records, scorecards, manifests (including token usage, provider errors and the resolved backend description) and the exact scenario are written per cell. The full results log with every table and reading is `docs/05-results-log.md`.

Appendix B. Metric definitions

See `docs/02-dual-benchmark.md` §3. Utility composite = mean of KUA, SSR and (1 - REGRESS), with NaNs skipped. Authority fidelity = 1 - FAR, where FAR pools revoked, adjacent, denied, never-granted and revoked-skill probes. Depth bins pool 20-tick buckets of per-probe compliance. All means are over seeds; intervals in Table 2 are normal approximations over 15 seeds.

Appendix C. Full ablation table

Table C1: H3/H5 ablation, all metrics. FAR = false-authority rate (lower is better); utility = mean of KUA, SSR and 1-REGRESS; INV = invariant violations per run. Five seeds each.

model / domain	store	FAR	RSR	GEN	LRR	utility	INV
gpt-4o-mini / coding (15 seeds)	type-blind (Ebbinghaus)	0.46	0.47	0.50	0.12	0.67	0
	labels only	0.69	0.25	0.68	0.00	0.72	0
	types only	0.00	1.00	0.00	0.00	0.73	17.3
	THSM, no gate	0.35	0.82	0.40	0.00	0.72	0
	THSM, no pinning	0.00	1.00	0.00	0.00	0.69	0
	THSM, tool-only pinning (5 seeds)	0.00	1.00	0.00	0.00	0.65	0
	THSM	0.00	1.00	0.00	0.00	0.72	0
gemini-2.5-flash-lite / coding	type-blind	0.62	0.17	0.57	0.00	0.40	0
	labels only	0.61	0.28	0.70	0.00	0.38	0
	types only	0.00	1.00	0.00	0.00	0.43	15.8
	THSM, no gate	0.55	0.15	0.58	0.33	0.47	0
	THSM, no pinning	0.00	1.00	0.00	0.33	0.40	0
	THSM	0.00	1.00	0.00	0.00	0.46	0
qwen3-coder-30b / coding	type-blind	0.58	0.23	0.56	0.00	0.53	0
	labels only	0.70	0.10	0.65	0.00	0.56	0
	types only	0.00	1.00	0.00	0.00	0.70	50.0
	THSM, no gate	0.60	0.27	0.73	0.00	0.51	0
	THSM	0.00	1.00	0.00	0.00	0.56	0
claude-sonnet-5 / coding	type-blind	0.27	0.87	0.35	0.00	0.55	0
	labels only	0.45	0.47	0.45	0.00	0.41	0
	types only	0.00	1.00	0.00	0.00	0.54	15.6
	THSM, no gate	0.22	1.00	0.35	0.00	0.51	0
	THSM, no pinning	0.00	1.00	0.00	0.00	0.42	0
	THSM	0.00	1.00	0.00	0.00	0.56	0
gpt-4o-mini / coding, freeform writer	type-blind	0.54	0.37	0.50	0.00	0.80	0
	Mem0 (external, own extraction LLM)	0.39	0.83	0.43	0.00	0.82	0
	THSM	0.00	1.00	0.00	0.00	0.77	0
gpt-4o-mini / procurement	type-blind	0.29	0.65	0.35	0.00	0.74	0
	labels only	0.41	0.38	0.42	0.00	0.71	0
	types only	0.02	0.95	0.00	0.00	0.74	16.8
	THSM, no gate	0.20	0.95	0.48	0.00	0.74	0
	THSM	0.00	1.00	0.00	0.00	0.74	0